Synthetic data quality: dilution and fragmentation

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Method	Dilution and relevance			Fragmentation	
	No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
Vanilla Cartridges	83.78	1.77	52.31	0.00	0.00
Question-guided (ours)	**46.69**	**98.17**	**91.06**	**23.66**	**78.72**
Δ guided − agnostic (pp)	-37.09	+96.41	+38.75	+23.66	+78.72

Source diagnostics n=993; query focus n=1000; cross-chunk diagnostics n=296. These are evidence-exposure diagnostics, not answer accuracy.

Synthetic data quality by document length

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Group	n_D	n_F	Method	Dilution and relevance			Fragmentation	
				No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
8K	249	60	Vanilla Cartridges	65.35	2.48	81.58	0.00	0.00
	249	60	Question-guided (ours)	**35.53**	**99.28**	**92.57**	**32.14**	**86.67**
16K	250	71	Vanilla Cartridges	83.57	2.03	64.49	0.00	0.00
	250	71	Question-guided (ours)	**44.83**	**98.57**	**95.20**	**20.61**	**81.69**
32K	247	81	Vanilla Cartridges	91.13	1.22	40.16	0.00	0.00
	247	81	Question-guided (ours)	**51.23**	**97.58**	**91.70**	**22.05**	**79.01**
64K	247	84	Vanilla Cartridges	95.23	1.34	22.65	0.00	0.00
	247	84	Question-guided (ours)	**55.29**	**97.28**	**84.72**	**21.71**	**70.24**

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.

Synthetic data quality by task

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Group	n_D	n_F	Method	Dilution and relevance			Fragmentation	
				No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
qa1	100	0	Vanilla Cartridges	87.32	0.93	43.00	—	—
	100	0	Question-guided (ours)	**36.91**	**98.74**	**99.00**	—	—
qa2	100	86	Vanilla Cartridges	81.08	0.33	53.00	0.00	0.00
	100	86	Question-guided (ours)	**35.95**	**99.36**	**89.50**	**25.99**	**82.56**
qa3	99	82	Vanilla Cartridges	78.41	3.97	53.54	0.00	0.00
	99	82	Question-guided (ours)	**54.18**	**99.72**	**77.10**	**17.02**	**57.32**
qa4	99	0	Vanilla Cartridges	88.33	2.16	46.46	—	—
	99	0	Question-guided (ours)	**65.74**	**94.70**	**72.73**	—	—
qa5	100	0	Vanilla Cartridges	87.18	1.88	60.00	—	—
	100	0	Question-guided (ours)	**55.52**	**95.53**	**88.00**	—	—
qa6	100	0	Vanilla Cartridges	88.23	1.65	49.00	—	—
	100	0	Question-guided (ours)	**54.41**	**99.94**	**97.00**	—	—
qa7	98	64	Vanilla Cartridges	76.23	1.85	56.62	0.00	0.00
	98	64	Question-guided (ours)	**28.83**	**98.66**	**95.83**	**26.82**	**90.62**
qa8	97	64	Vanilla Cartridges	77.50	1.44	51.55	0.00	0.00
	97	64	Question-guided (ours)	**35.50**	**95.64**	**96.39**	**25.85**	**89.06**
qa9	100	0	Vanilla Cartridges	86.92	2.47	56.00	—	—
	100	0	Question-guided (ours)	**50.75**	**99.88**	**98.00**	—	—
qa10	100	0	Vanilla Cartridges	86.26	1.01	54.00	—	—
	100	0	Question-guided (ours)	**48.67**	**99.58**	**97.00**	—	—

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.

Synthetic data quality: all tasks at 8K

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Group	n_D	n_F	Method	Dilution and relevance			Fragmentation	
				No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
qa1	25	0	Vanilla Cartridges	67.56	1.57	68.00	—	—
	25	0	Question-guided (ours)	**38.23**	**100.00**	**100.00**	—	—
qa2	25	19	Vanilla Cartridges	62.07	0.69	76.00	0.00	0.00
	25	19	Question-guided (ours)	**22.43**	**99.83**	**86.00**	**36.57**	**84.21**
qa3	24	13	Vanilla Cartridges	60.16	7.77	**81.94**	0.00	0.00
	24	13	Question-guided (ours)	**28.84**	**100.00**	76.39	**32.38**	**69.23**
qa4	25	0	Vanilla Cartridges	76.13	2.42	56.00	—	—
	25	0	Question-guided (ours)	**43.29**	**99.60**	**96.00**	—	—
qa5	25	0	Vanilla Cartridges	72.26	1.25	**92.00**	—	—
	25	0	Question-guided (ours)	**58.01**	**97.62**	80.00	—	—
qa6	25	0	Vanilla Cartridges	74.61	2.54	84.00	—	—
	25	0	Question-guided (ours)	**47.88**	**99.88**	**100.00**	—	—
qa7	25	12	Vanilla Cartridges	49.33	1.25	93.20	0.00	0.00
	25	12	Question-guided (ours)	**18.00**	**98.75**	**98.67**	**25.00**	**91.67**
qa8	25	16	Vanilla Cartridges	48.98	1.50	92.67	0.00	0.00
	25	16	Question-guided (ours)	**28.93**	**97.25**	**100.00**	**32.05**	**100.00**
qa9	25	0	Vanilla Cartridges	71.93	3.50	92.00	—	—
	25	0	Question-guided (ours)	**40.47**	**100.00**	**100.00**	—	—
qa10	25	0	Vanilla Cartridges	70.23	2.32	80.00	—	—
	25	0	Question-guided (ours)	**28.94**	**99.83**	**88.00**	—	—

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.

Synthetic data quality: all tasks at 16K

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Group	n_D	n_F	Method	Dilution and relevance			Fragmentation	
				No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
qa1	25	0	Vanilla Cartridges	91.30	0.60	56.00	—	—
	25	0	Question-guided (ours)	**31.83**	**98.95**	**100.00**	—	—
qa2	25	21	Vanilla Cartridges	83.04	0.12	52.00	0.00	0.00
	25	21	Question-guided (ours)	**34.23**	**98.50**	**94.00**	**21.83**	**85.71**
qa3	25	21	Vanilla Cartridges	75.62	3.75	68.00	0.00	0.00
	25	21	Question-guided (ours)	**58.25**	**99.62**	**84.00**	**15.48**	**61.90**
qa4	25	0	Vanilla Cartridges	89.18	1.88	56.00	—	—
	25	0	Question-guided (ours)	**53.06**	**95.12**	**92.00**	—	—
qa5	25	0	Vanilla Cartridges	85.83	3.25	80.00	—	—
	25	0	Question-guided (ours)	**51.71**	**96.38**	**92.00**	—	—
qa6	25	0	Vanilla Cartridges	85.50	3.27	64.00	—	—
	25	0	Question-guided (ours)	**55.22**	**100.00**	**96.00**	—	—
qa7	25	15	Vanilla Cartridges	77.54	1.50	66.20	0.00	0.00
	25	15	Question-guided (ours)	**27.33**	**99.12**	**96.00**	**23.33**	**93.33**
qa8	25	14	Vanilla Cartridges	77.00	1.50	62.67	0.00	0.00
	25	14	Question-guided (ours)	**31.46**	**98.25**	**98.00**	**23.59**	**92.86**
qa9	25	0	Vanilla Cartridges	85.62	3.50	68.00	—	—
	25	0	Question-guided (ours)	**54.29**	**99.88**	**100.00**	—	—
qa10	25	0	Vanilla Cartridges	85.04	0.96	72.00	—	—
	25	0	Question-guided (ours)	**50.88**	**99.88**	**100.00**	—	—

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.

Synthetic data quality: all tasks at 32K

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Group	n_D	n_F	Method	Dilution and relevance			Fragmentation	
				No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
qa1	25	0	Vanilla Cartridges	94.50	1.04	24.00	—	—
	25	0	Question-guided (ours)	**34.26**	**100.00**	**100.00**	—	—
qa2	25	22	Vanilla Cartridges	87.55	0.38	52.00	0.00	0.00
	25	22	Question-guided (ours)	**42.12**	**99.50**	**92.00**	**26.51**	**86.36**
qa3	25	24	Vanilla Cartridges	85.62	2.62	40.00	0.00	0.00
	25	24	Question-guided (ours)	**58.88**	**99.62**	**76.00**	**14.58**	**54.17**
qa4	24	0	Vanilla Cartridges	92.19	1.72	45.83	—	—
	24	0	Question-guided (ours)	**75.00**	**93.33**	**70.83**	—	—
qa5	25	0	Vanilla Cartridges	94.12	1.75	40.00	—	—
	25	0	Question-guided (ours)	**52.38**	**94.62**	**88.00**	—	—
qa6	25	0	Vanilla Cartridges	94.79	0.42	36.00	—	—
	25	0	Question-guided (ours)	**53.67**	**100.00**	**100.00**	—	—
qa7	25	18	Vanilla Cartridges	86.62	1.00	42.07	0.00	0.00
	25	18	Question-guided (ours)	**33.46**	**97.62**	**96.00**	**28.70**	**100.00**
qa8	23	17	Vanilla Cartridges	90.47	1.75	33.33	0.00	0.00
	23	17	Question-guided (ours)	**49.17**	**92.29**	**97.83**	**19.77**	**82.35**
qa9	25	0	Vanilla Cartridges	92.14	1.12	48.00	—	—
	25	0	Question-guided (ours)	**59.59**	**99.75**	**96.00**	—	—
qa10	25	0	Vanilla Cartridges	93.25	0.38	40.00	—	—
	25	0	Question-guided (ours)	**54.54**	**99.00**	**100.00**	—	—

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.

Synthetic data quality: all tasks at 64K

Matched Qwen3-4B BABILong synthetic datasets; 27,949 examples per method

Group	n_D	n_F	Method	Dilution and relevance			Fragmentation	
				No-support ↓	Query focus ↑	Support recall ↑	Joint exercise ↑	Joint case ↑
qa1	25	0	Vanilla Cartridges	95.92	0.50	24.00	—	—
	25	0	Question-guided (ours)	**43.33**	**96.00**	**96.00**	—	—
qa2	25	24	Vanilla Cartridges	91.66	0.12	32.00	0.00	0.00
	25	24	Question-guided (ours)	**45.02**	**99.62**	**86.00**	**20.79**	**75.00**
qa3	25	24	Vanilla Cartridges	91.50	1.75	25.33	0.00	0.00
	25	24	Question-guided (ours)	**69.75**	**99.62**	**72.00**	**12.50**	**50.00**
qa4	25	0	Vanilla Cartridges	95.98	2.64	28.00	—	—
	25	0	Question-guided (ours)	**91.97**	**90.75**	**32.00**	—	—
qa5	25	0	Vanilla Cartridges	96.50	1.25	28.00	—	—
	25	0	Question-guided (ours)	**60.00**	**93.50**	**92.00**	—	—
qa6	25	0	Vanilla Cartridges	98.00	0.38	12.00	—	—
	25	0	Question-guided (ours)	**60.89**	**99.88**	**92.00**	—	—
qa7	23	19	Vanilla Cartridges	92.75	3.67	22.25	0.00	0.00
	23	19	Question-guided (ours)	**37.18**	**99.12**	**92.39**	**28.95**	**78.95**
qa8	24	17	Vanilla Cartridges	95.31	1.00	14.58	0.00	0.00
	24	17	Question-guided (ours)	**33.46**	**94.75**	**89.58**	**27.94**	**82.35**
qa9	25	0	Vanilla Cartridges	98.00	1.75	16.00	—	—
	25	0	Question-guided (ours)	**48.65**	**99.88**	**96.00**	—	—
qa10	25	0	Vanilla Cartridges	96.50	0.38	24.00	—	—
	25	0	Question-guided (ours)	**60.33**	**99.62**	**100.00**	—	—

Source-mapped cases: nD; cross-chunk subset: nF; focus uses all planned cases. Gray: vanilla; cyan: guided; red: better; underline: second.